Jammu Smart City Hackathon 2026 · Digital Governance, Mobility & Smart Infrastructure

ParkPulse Jammu

Prototype screens, wireframes and supporting documents

Live parking and congestion intelligence from the city's existing cameras: free-slot counts for citizens and pilgrims on WhatsApp, automatic no-parking alerts for traffic police, and forecasts and hotspots for JMC and JSCL.

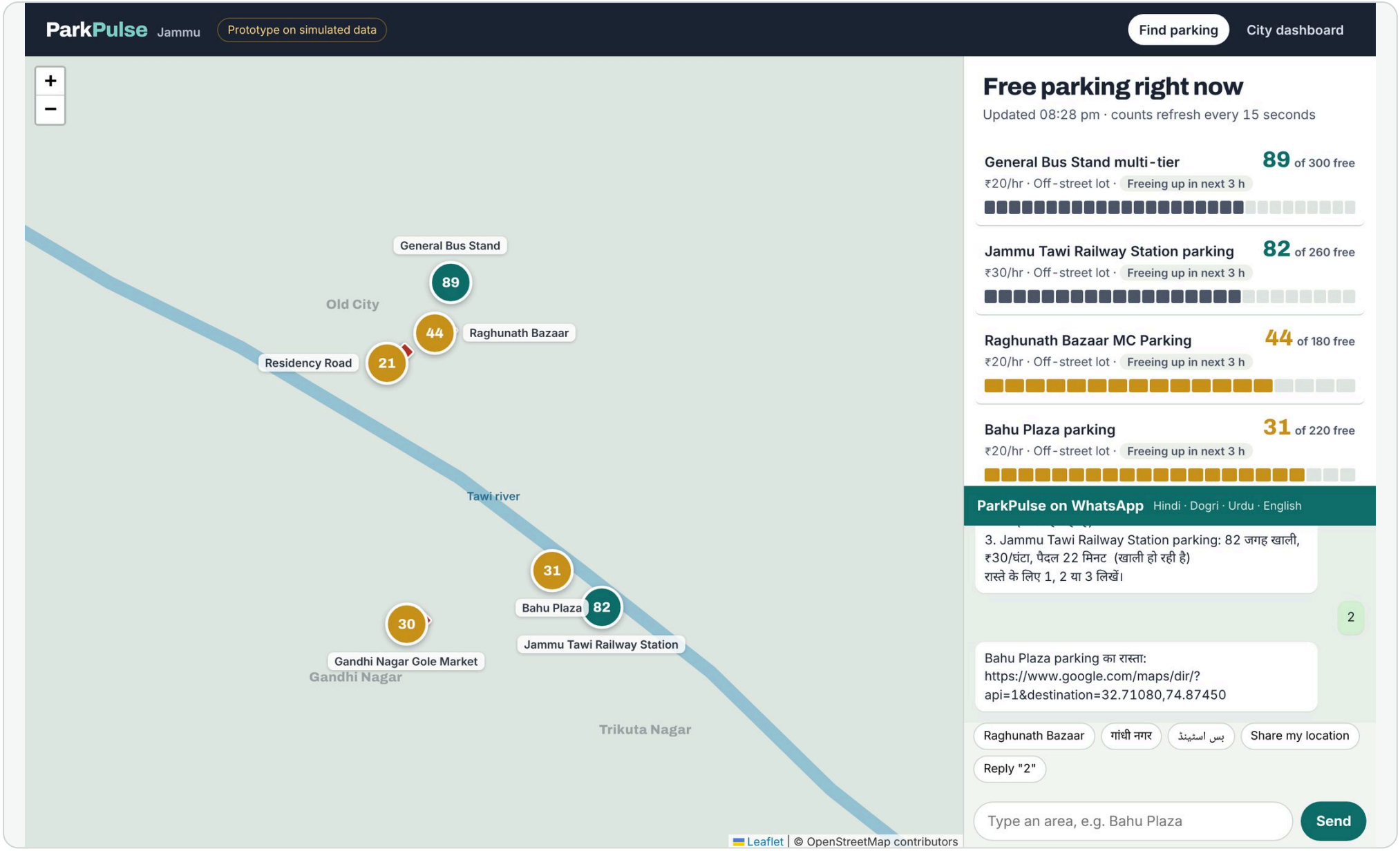
What you are looking at
Screens captured from the working prototype in the GitHub repository. Every screen is live code, not a mock-up.

What is simulated
Occupancy, violations and fee figures come from a simulator, and camera frames are synthetic. No Jammu feed or survey data has been used yet.

Code
github.com/Aditya-eng/ParkPulse
Run: pip install -r requirements.txt, then uvicorn parkpulse.app:app

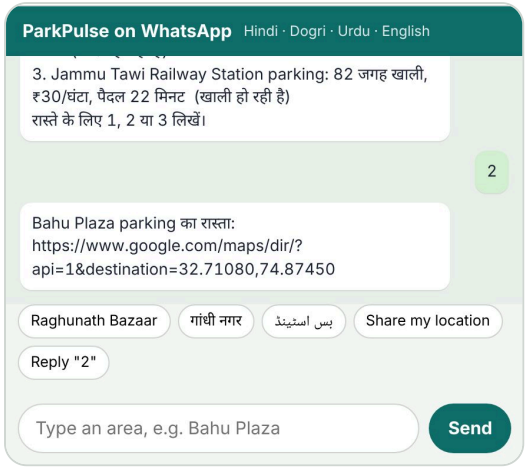
Citizen view: where can I park right now?

Each zone shows free bays as a number and as a strip of painted bays that fills as the lot fills. Colour changes from teal to mustard to red as a zone gets busy. Tapping a zone flies the map to it and offers directions.

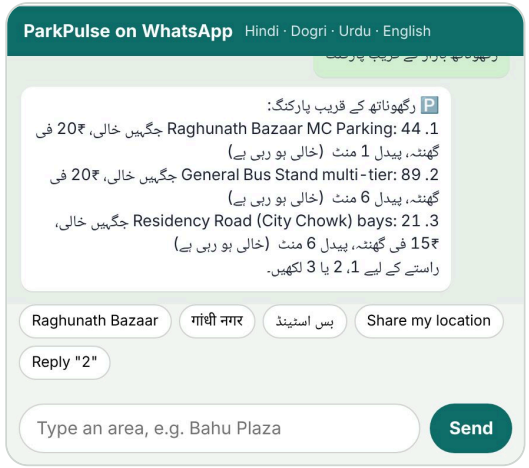


WhatsApp bot in four languages

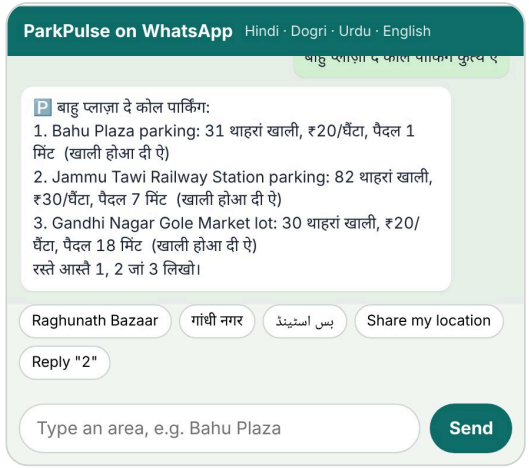
No app download. People type an area in Hindi, Dogri, Urdu or English (typos are fine), or share a location pin. The bot lists the three nearest zones with free bays, fee and walking time, and replying 1, 2 or 3 returns directions.



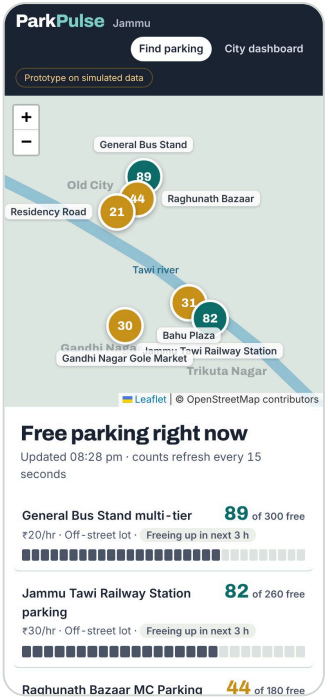
Hindi query, then "2" for directions to Bahu Plaza.



Urdu query for Raghunath Bazaar, answered right to left.



Dogri query. Dogri and Urdu copy is a first draft for native-speaker review.



The same view on a phone browser.

City dashboard for JMC and JSCL

Citywide totals, a 6-hour forecast per zone, an hour-by-hour fill pattern for every zone, live no-parking alerts, enforcement hotspots and an estimate of uncollected fees.



Reading the dashboard

The heatmap shows market zones filling at 12–13h and again at 18–19h, while the bus stand and railway station peak at 8–9h. That is where marshals and overflow lots help most.

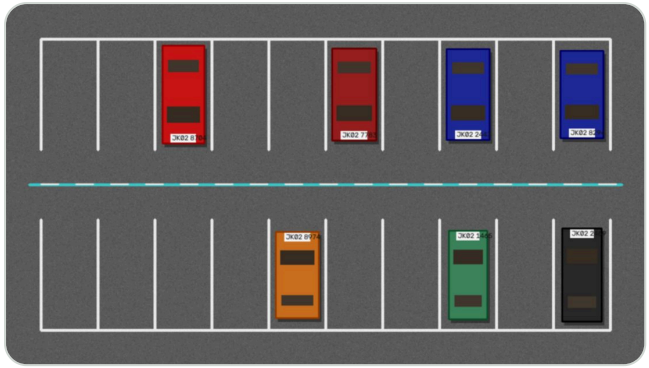
The forecast line warns before a zone crosses the 95% "full" line, so fees or marshals can be adjusted hours ahead.

Hotspots rank no-parking segments by alerts over 14 days, so enforcement time goes where it changes most.

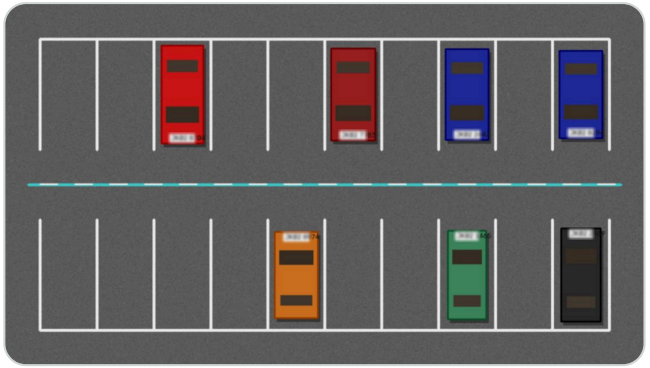
> The fee estimate assumes 18% of occupied bay-hours go unbilled. It is a placeholder to be replaced with JMC billing data in a pilot.

Vision pipeline on a camera frame

The edge device reads a camera frame, blurs plates and faces, and counts each painted slot as free or taken. Only the counts leave the device in normal operation.



1. Frame from camera (synthetic top-down lot).



2. Plates blurred before storage or display.



3. Slots marked free (green) or taken (red): 13 of 20 free.

Measured on 50 synthetic frames (1,000 slots)

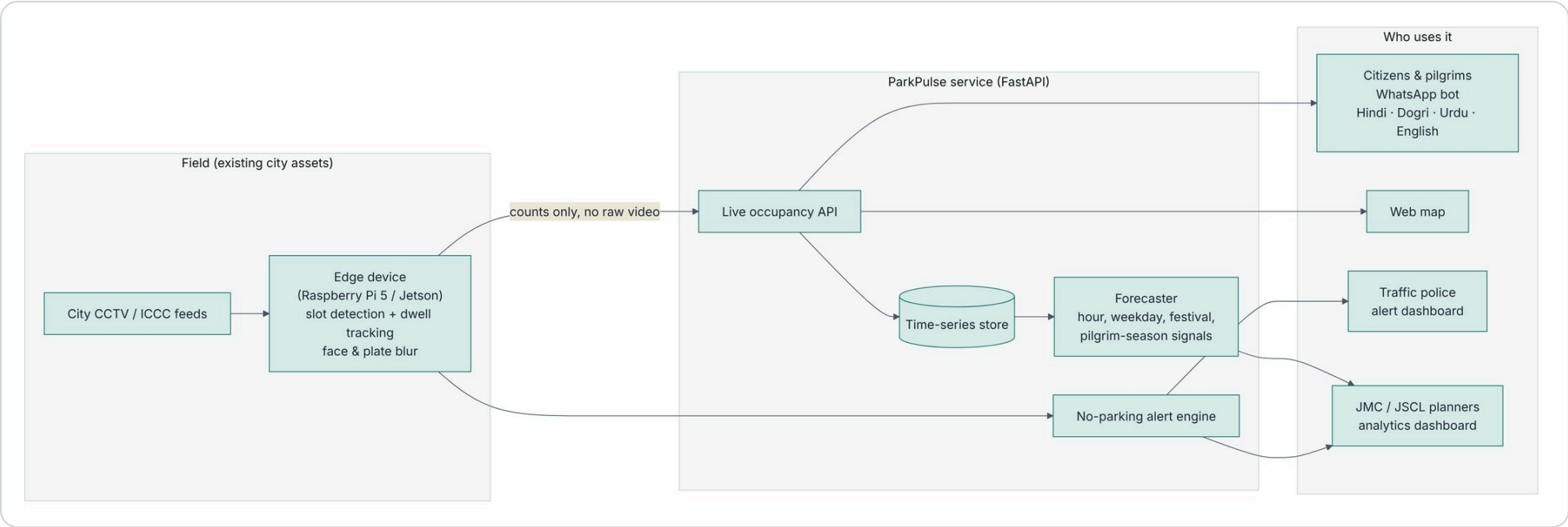
Check	Result
Slot free / taken accuracy	**100%**
Number plates found and blurred	89%
Illegal-parking alert after 5-minute dwell	**Raised once per vehicle**

What changes with real cameras

Synthetic frames are easy, so 100% here says the pipeline works, not that it is ready. On Jammu CCTV, slot polygons are drawn per camera with the included calibration tool, and oblique angles switch to the YOLO vehicle-detection backend. A trained Indian-plate detector replaces the plate heuristic before any deployment.

How the pieces fit

Existing CCTV feeds an edge device per site. A FastAPI service keeps live occupancy, runs forecasts and alerts, and serves the WhatsApp bot, the web map and the city dashboard.



Edge

Raspberry Pi 5 or Jetson Orin Nano. OpenCV slot detector by default, optional YOLO. Face and plate blurring on the device.

Service

Python, FastAPI, scikit-learn gradient boosting. Time-series store (TimescaleDB in a pilot). Docker image included.

People

WhatsApp Cloud API webhook, Leaflet web map, Chart.js dashboard. Works offline for demos with vendored libraries.

Forecast quality and pilot plan

The forecaster was back-tested on the last 7 days of 8 weeks of simulated history and compared with the simple rule "same hour last week".

Zone	ParkPulse error	Same hour last week
Gandhi Nagar Gole Market lot	**6.1 pts**	8.0 pts
Raghunath Bazaar MC Parking	**7.1 pts**	8.7 pts
Residency Road (City Chowk) bays	**7.1 pts**	9.5 pts
Jammu Tawi Railway Station parking	**7.9 pts**	9.5 pts
Bahu Plaza parking	**8.0 pts**	9.1 pts
General Bus Stand multi-tier	**9.1 pts**	10.3 pts

Mean absolute error in occupancy percentage points. Simulated history, so this shows the method works; real error is measured in the pilot.

Pilot targets, to be measured: 25–30% less time spent searching for parking, 20–25% fewer illegal-parking incidents, and 15–20% higher fee collection in pilot zones. Prototype cost for one zone: about ₹75,000, or about ₹20,000 if existing city camera feeds are shared.

12-week pilot

Wk 1–2	Choose 2–3 zones with JMC (Raghunath Bazaar, Residency Road, Gandhi Nagar). Get camera feed access or mount 2–3 low-cost cameras.
Wk 3–4	Draw slot polygons, label real frames, choose detector backend per camera, check against manual counts.
Wk 5–8	Run the bot with a test group, route alerts to one traffic police unit, review Dogri and Urdu copy with native speakers.
Wk 9–12	Measure search time, illegal-parking incidents and fee collection against the pre-pilot baseline; decide on ward-by-ward rollout.

What we need from the city

Read access to camera feeds for pilot zones, current parking fee and capacity records, and a traffic police contact for alert routing.